

Lab Assignment 2

CS 331: Machine Learning

Instructor: Dr. Divya Padmanabhan, Dr. Satyanath Bhat

Teaching Assistants: Vipin, Tina, Manvi, Nimra, Sunny

To be submitted by 5 pm on January 20, 2025

You will explore the Bayes classifier for Normal distributions (univariate as well as multivariate normal) in this lab. Recall that the density of a multivariate Gaussian $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is,

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{(\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}{2} \right)$$

Use the following distributions for the questions that follow.

$p_0 = 0.5, p_1 = 0.5, f([X_1, X_2, X_3] | Y = 0) \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), f([X_1, X_2, X_3] | Y = 1) \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$

1. Generate a dataset of n points. You may use python Numpy package. The mean vectors, covariance matrix and n should be taken as input.
2. Using a single feature X_i at a time, implement Bayes Classifier and compute empirical classification error (0-1 loss).
3. Using two features X_i and X_j at a time, implement Bayes Classifier and compute empirical classification error (0-1 loss).
4. Using all features X_1, X_2 and X_3 , implement Bayes Classifier and compute empirical classification error (0-1 loss).
5. Do the errors increase/decrease when you use more features? In order to check this you need to generate several datasets of n points (where n is significantly large) and check the difference in errors.
6. For all the cases (single feature, 2 features, 3 features), implement a classifier which assigns the class with the closest mean to the feature vector. Compute empirical classification error (0-1 loss).

Points to pay attention to:

- For questions 2,3,4 you need to assign a class with maximum value of $p_i f_i(\mathbf{x})$. You may make use of built in functions in numpy/scipy to calculate the multivariate normal density function.
- You will benefit immensely by modularizing your code, since there are a total of at least 14 classifiers to be implemented!
- For the case where $\boldsymbol{\Sigma}_0 = \boldsymbol{\Sigma}_1 = \sigma^2 I$, how do each of the classifiers in 2,3,4 compare with the corresponding classifier in 5 that acts on the same feature set?